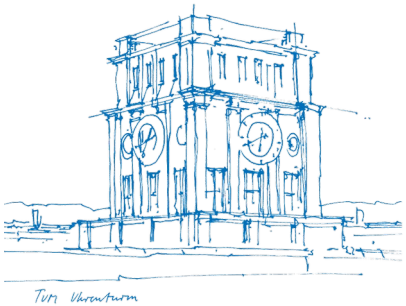
On the Feasibility of Pixel-Free Visual Odometry Using Video Codec Motion Vectors

Emir Özer

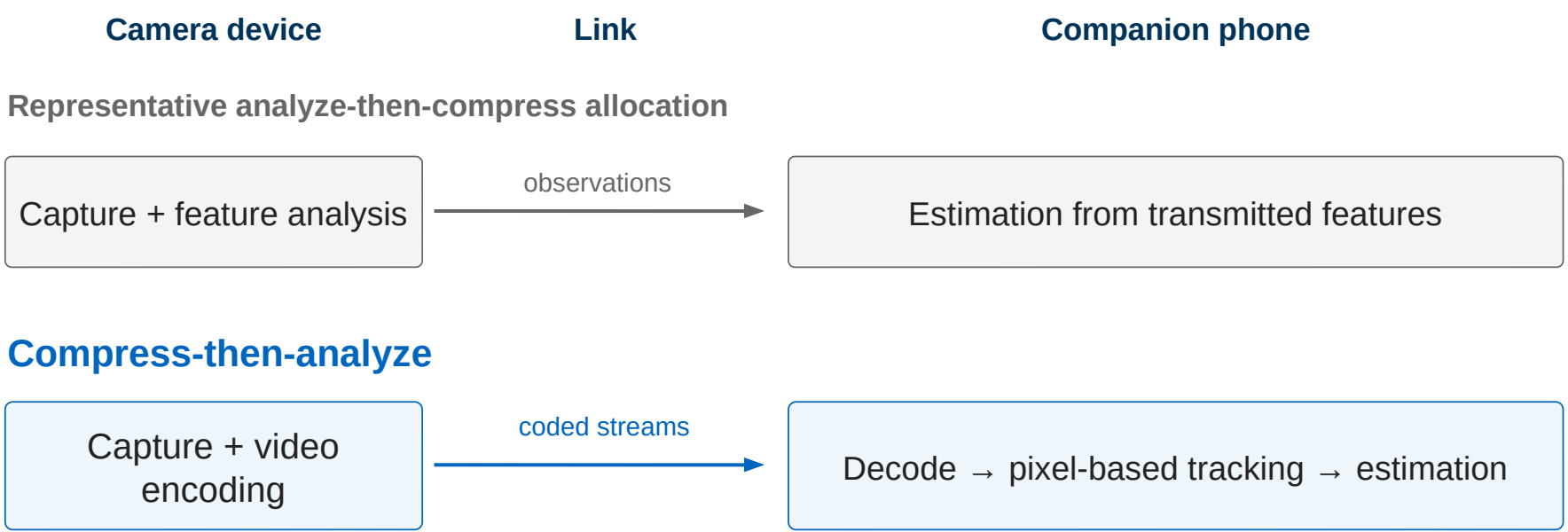
Master’s Thesis in Informatics

Supervisor: Mateo de Mayo
Examiner: Daniel Cremers

28 September 2026



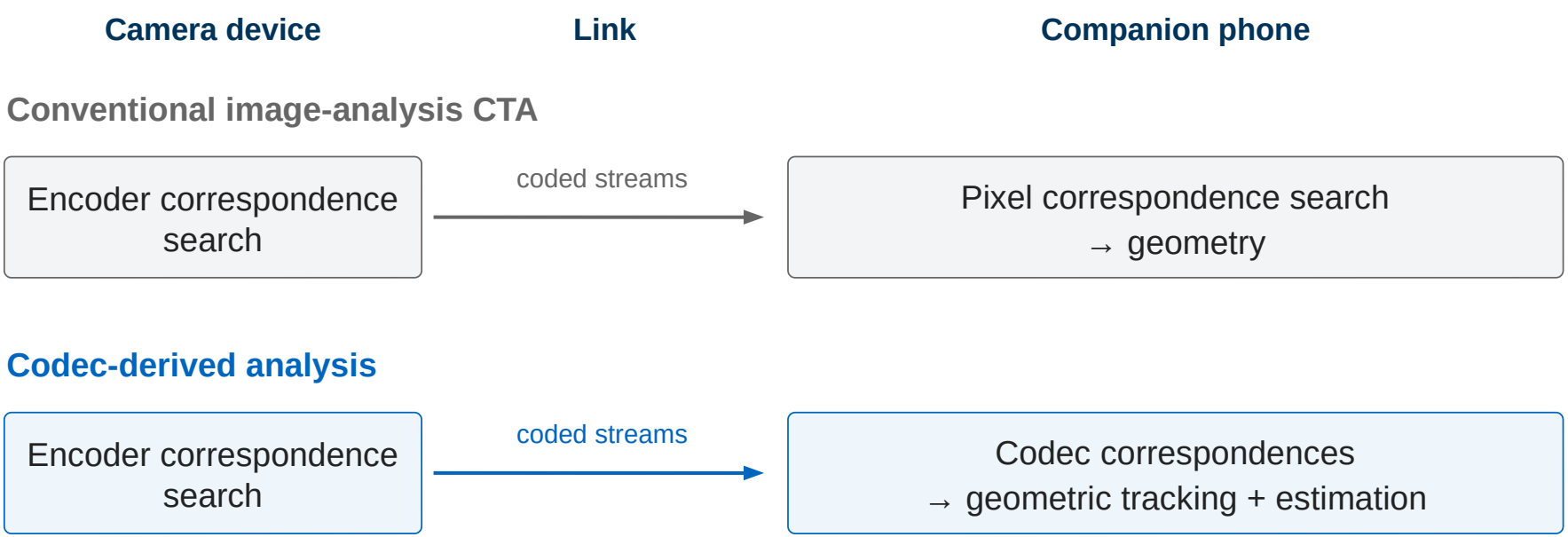
First opportunity: place analysis on the receiver



Move analysis off the constrained camera.

CTA/ATC motivation: thesis §§1.1, 3.1; Hofer et al. [2], Redondi et al. [3].

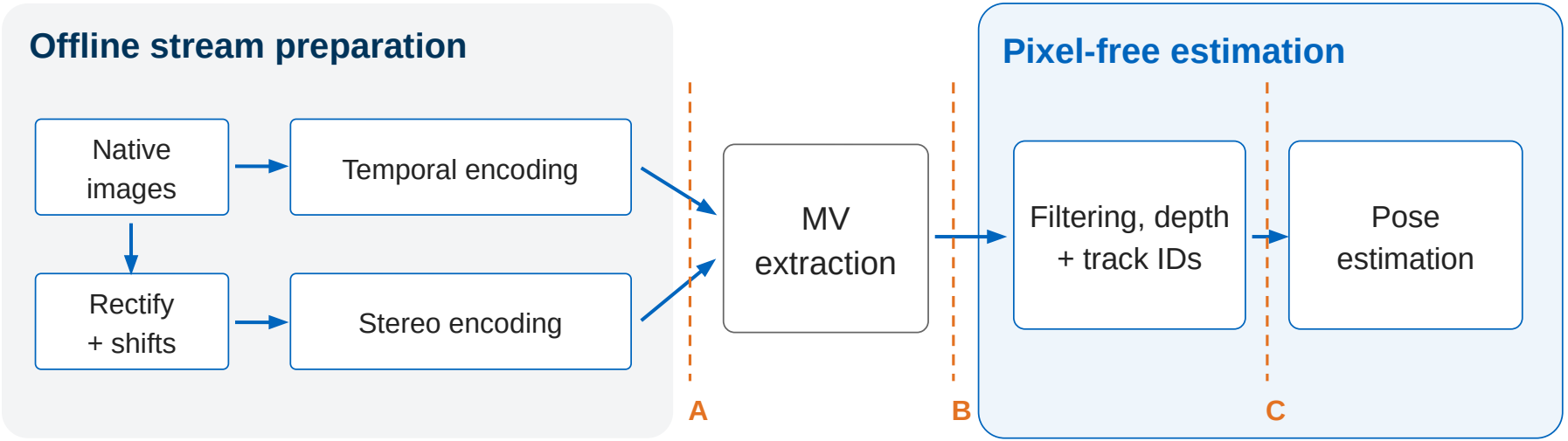
Second opportunity: reuse encoding work



Start geometric tracking from correspondences the encoder supplies.

Codec reuse motivation: thesis §§1.1, 7.6; Chaurasia et al. [1]. Representative architectures, not measured efficiency gains.

One pipeline, three different boundaries

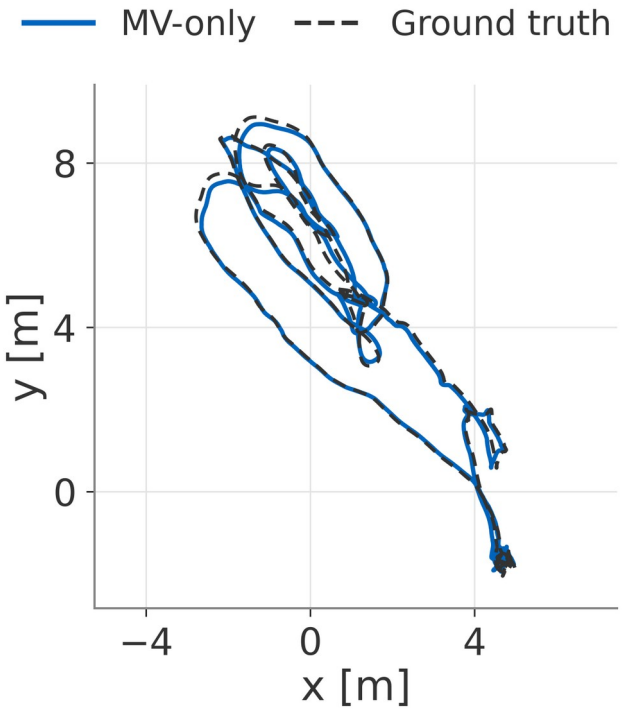


Sequential estimation from prepared inputs • calibrated stereo • no IMU

Possible transmission cuts: A coded streams B observations C confirmed tracks

Thesis §§1.2, 4.1–4.2, 7.6. Transport cuts are prospective; the prototype uses prepared streams.

Codec motion can recover a camera trajectory



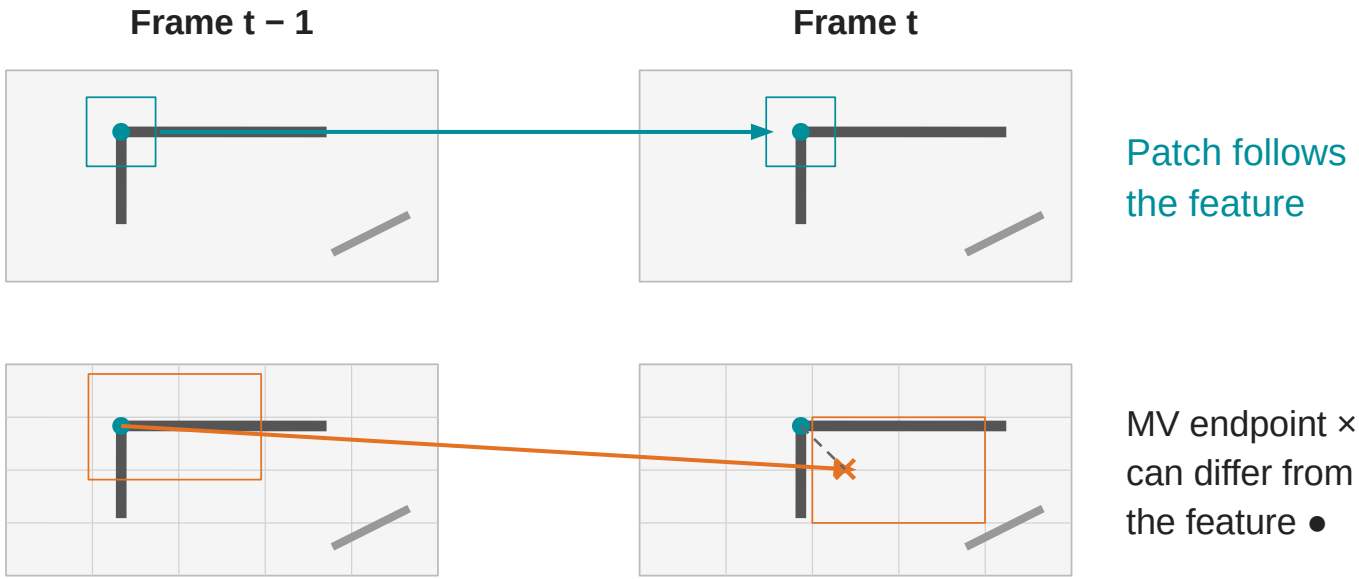
EMH01
EuRoC Machine Hall

Metric tracking from motion vectors

No image refinement
in tracking or estimation

EMH01 · complete saved trajectory · fixed-scale rigid alignment. Thesis Table 6.4; variant documented in notes.

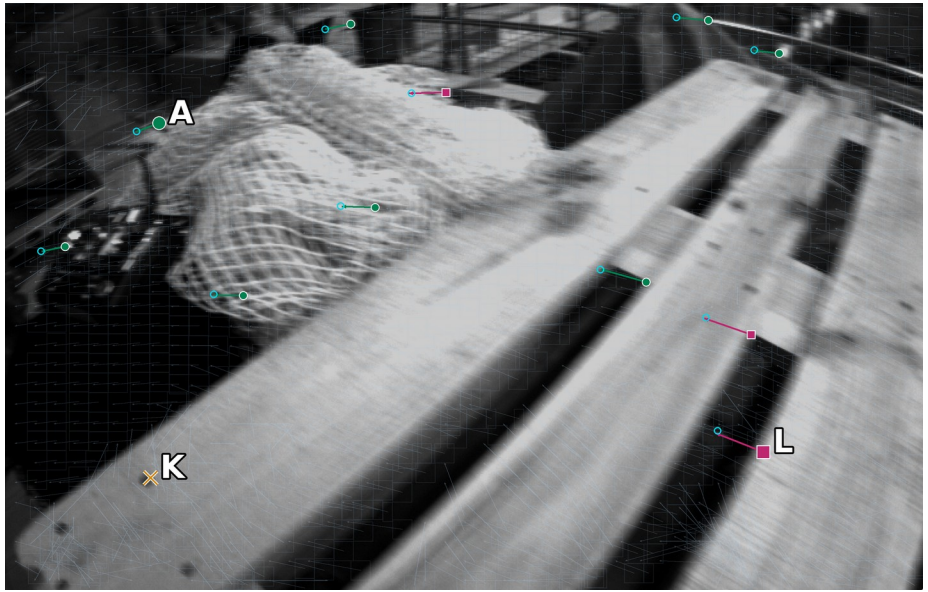
A block prediction proposes motion, not point identity



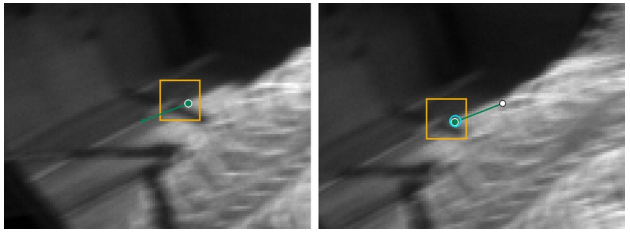
Block support is not a persistent feature identity.

Many motion candidates; different kinds of support

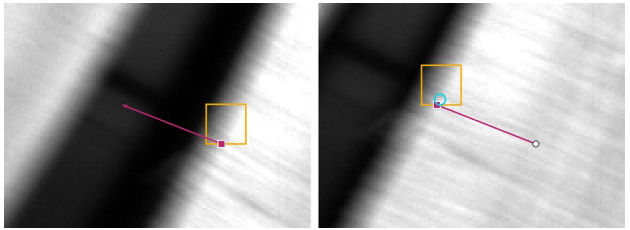
● **Close to KLT (≤1.5 px)** ■ **Disagrees (>1.5 px)** × **No covering source support**



A · gap 0.39 px Source → current



L · gap 2.47 px Source → current

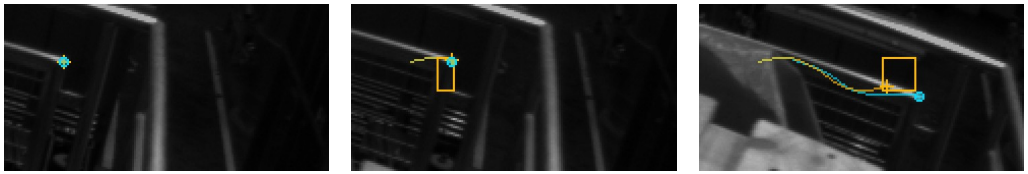


Source view + forward displacements; cyan rings: KLT endpoints.

Post-hoc image features / KLT are diagnostics, never PVO inputs. Source-support containment determines which vector can be applied.

A continuing vector is not a continuing identity

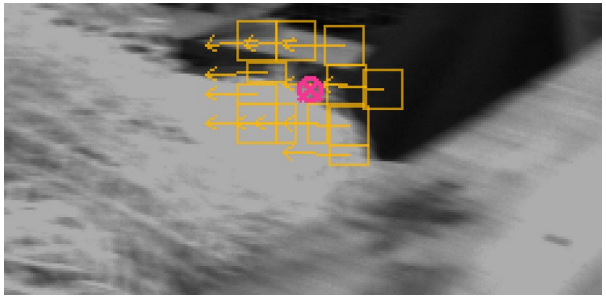
Vectors present; point identity drifts



Initialization | 4 steps · 0.86 px | 14 steps · 16.28 px

Orange: raw MV propagation Cyan: KLT diagnostic

The feature remains; support is lost



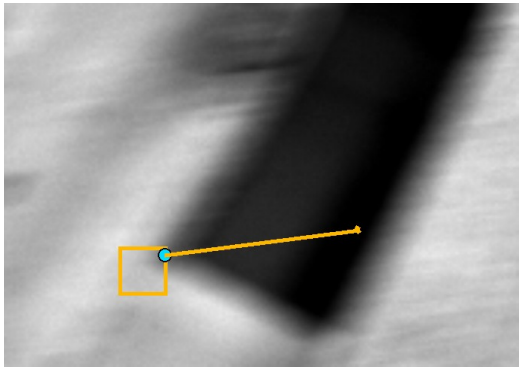
Magenta ×: unsupported query

Nearby vectors cannot replace a missing covering source block.

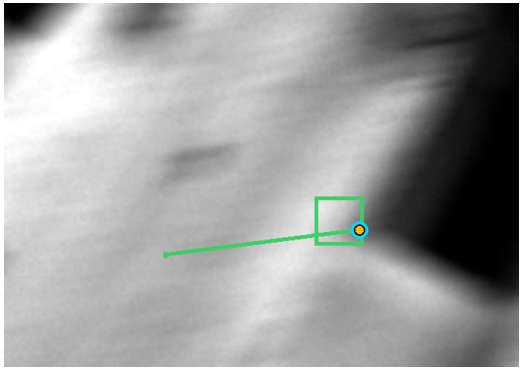
Separate measured EMH01 examples · thesis Figs. 4.7–4.8. KLT is an image-tracking diagnostic, not ground truth.

First establish reciprocal temporal support

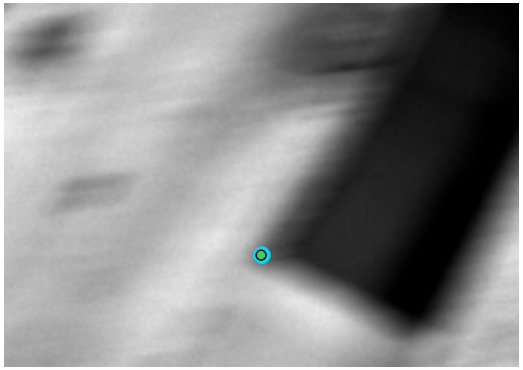
Forward: 64.78 px



Reverse: 65.03 px



Cycle closure: 0.25 px

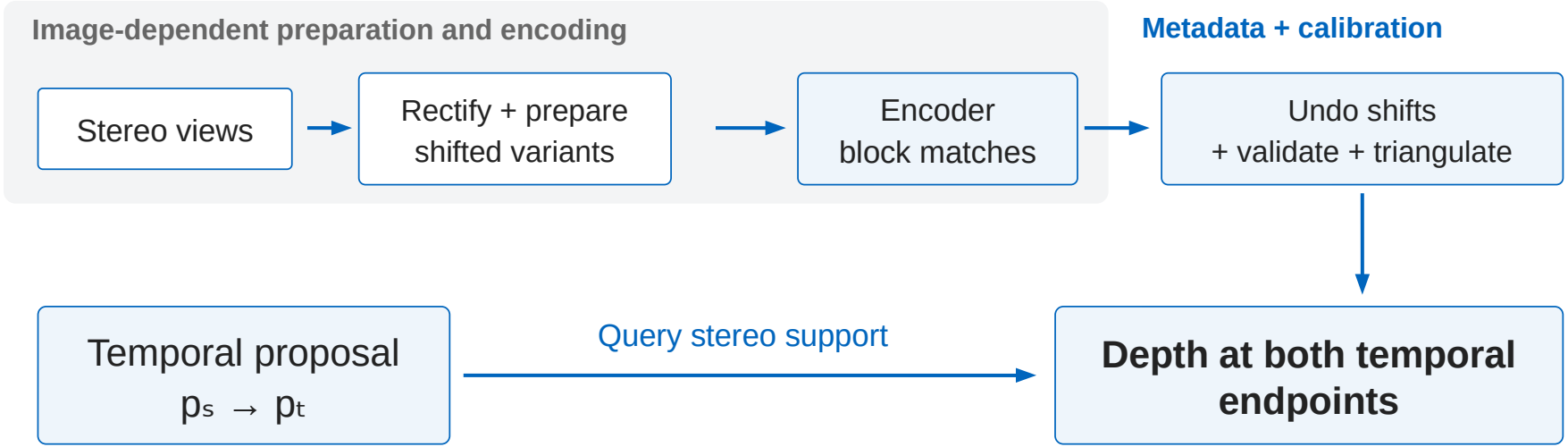


Median consistency → compatible directions → cycle closure + overlapping supports

Provisional birth at t → Reciprocal third view confirms ID

Measured EMH01 example · thesis Fig. 4.9 / §§4.5–4.6. Independent reverse encoding is prepared offline.

Stereo turns temporal proposals into metric geometry



One temporal displacement becomes one metric 3D pair.

Stereo overview · thesis §§4.3–4.4, 4.7. Preparation uses pixels; lookup and triangulation use metadata and calibration.

Use left–right–left encoding for stereo

All three pictures come from the same synchronized stereo pair.

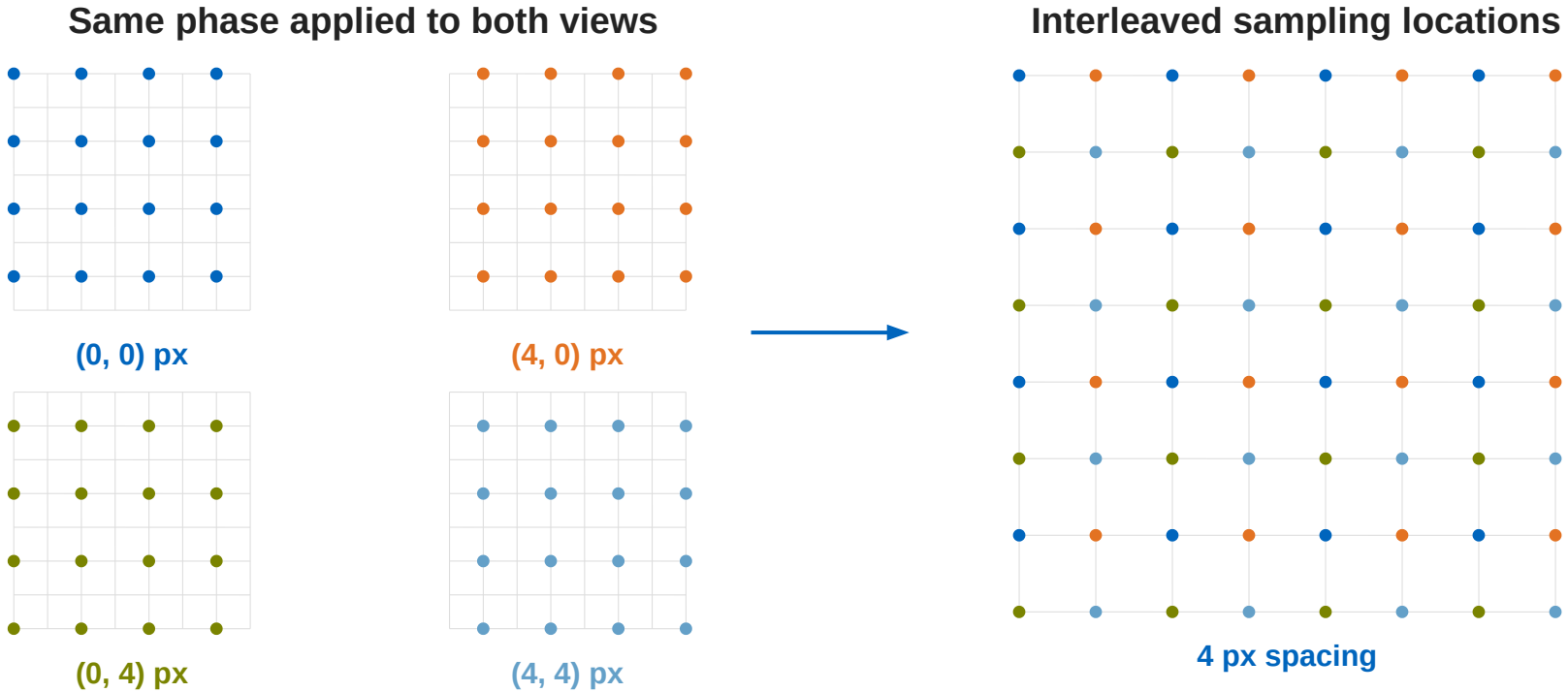


Opposite stereo directions from the two prediction steps

Restore coordinates → calibrated geometric checks → depth

Encoder stereo adapted from Chaurasia et al. [1] · thesis §4.4. Independent I–P–P triples; ref=1.

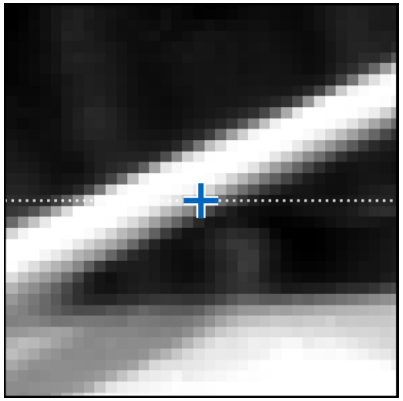
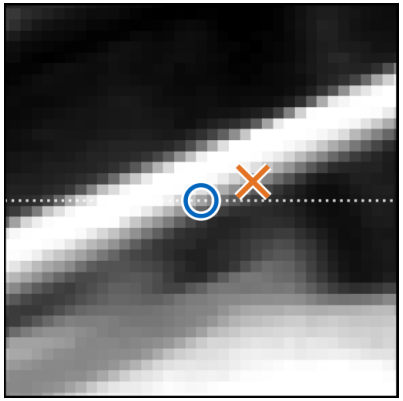
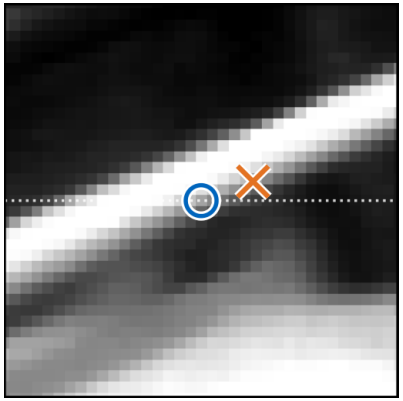
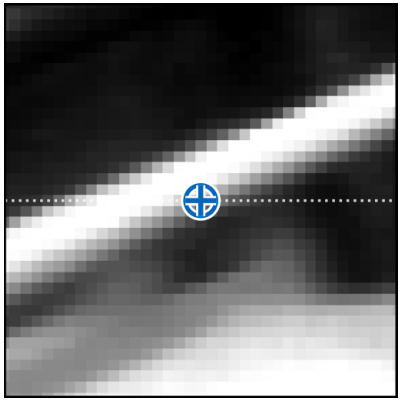
Grid phases improve spatial coverage



Schematic of thesis Fig. 4.2 · disparity is preserved. 4-pixel sampling does not mean every encoded block is 4×4.

A pre-shift makes a nearby match accessible

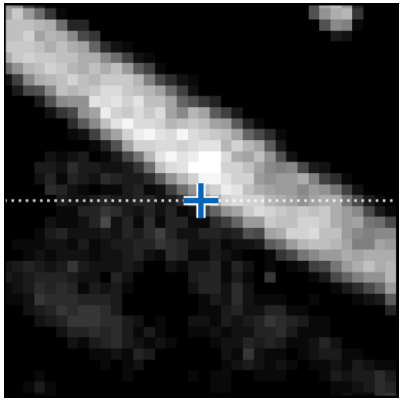
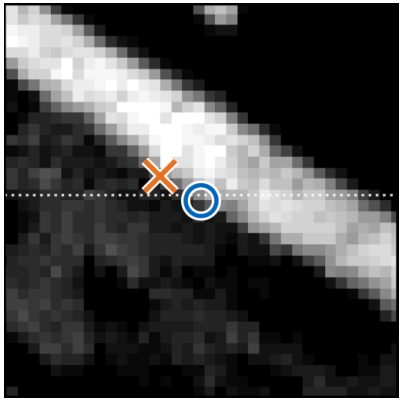
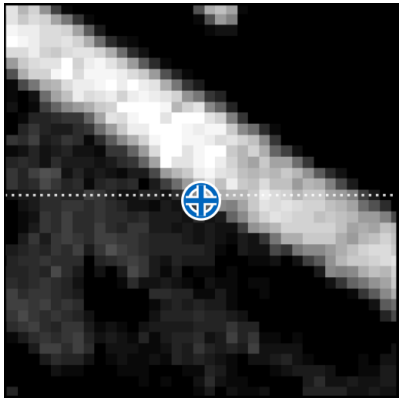
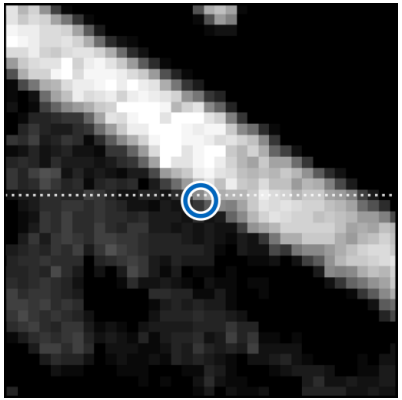
EMH01 · 135 s · Z = 1.31 m · grid phase (4,4) px

Query	s = 0 px	s = 8 px	s = 32 px
			
Disparity: 29.25 px	**No match**	**No match**	**Match**
Same query	Vertical error: 1.75 px	Vertical error: 1.75 px	Accepted proposal
|d − s| [px]	**29.25**	**21.25**	**2.75**
Temporal proposals	**Stereo geometry**	Track association	Pose estimation

Thesis Fig. 4.3. Exact source panels and markers. Pre-shifts give complementary match availability, not guaranteed depth bins.

A different depth benefits from a different shift

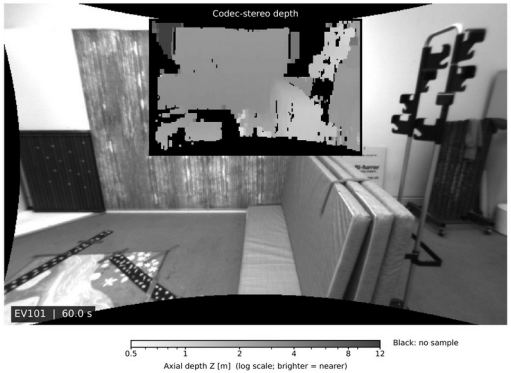
EMH02 · 95 s · Z = 4.52 m · grid phase (4,0) px

Query	s = 0 px	s = 8 px	s = 32 px
			
Disparity: 8.50 px	**No match**	**Match**	**No match**
Same query	Vertical error: 1.75 px	Accepted proposal	No covering block
|d − s| [px]	**8.50**	**0.50**	**23.50**
Temporal proposals	**Stereo geometry**	Track association	Pose estimation

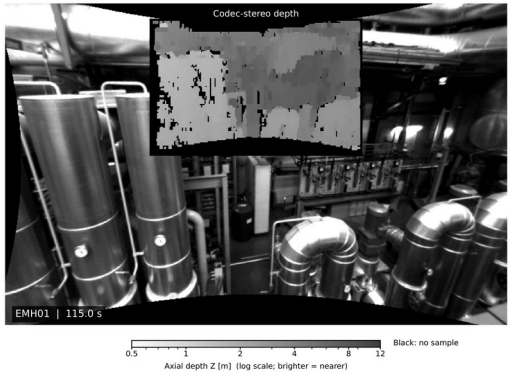
Thesis Fig. 4.3. Exact source panels and markers. Pre-shifts give complementary match availability, not guaranteed depth bins.

Recovered depth supports both temporal endpoints

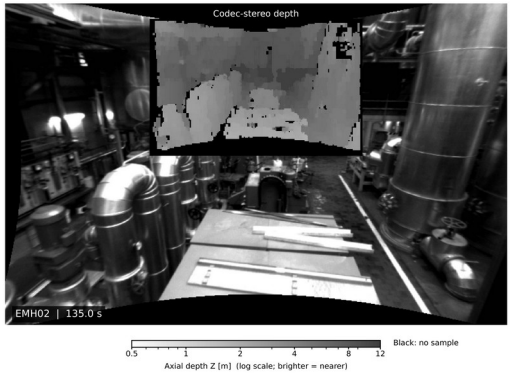
EV101 · 60 s



EMH01 · 115 s



EMH02 · 135 s



Temporal endpoint p_s → Temporal endpoint p_t → **Metric pair: P_s, P_t**

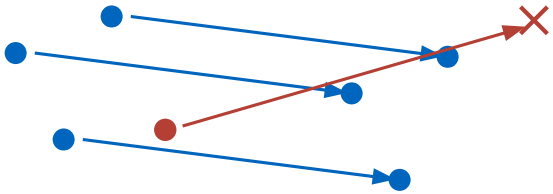
Temporal proposals | **Stereo geometry** | Track association | Pose estimation

Thesis Figs. 4.4–4.5 · brighter is nearer; black is unsupported. Display: rectified axial depth; estimator: native-ray ranges.

Turn candidate motion into geometric tracks

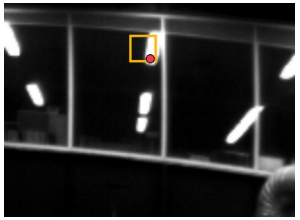
Actual codec candidates → Stereo depth at both endpoints → Putative 3D pairs → TEASER++ rigid consensus

Pairs share one rigid motion (schematic)

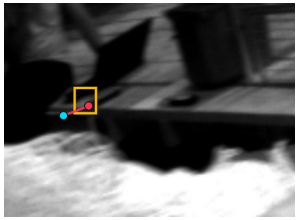


Blue: consensus Red ×: inconsistent pair

Sourced geometric decisions, separate examples

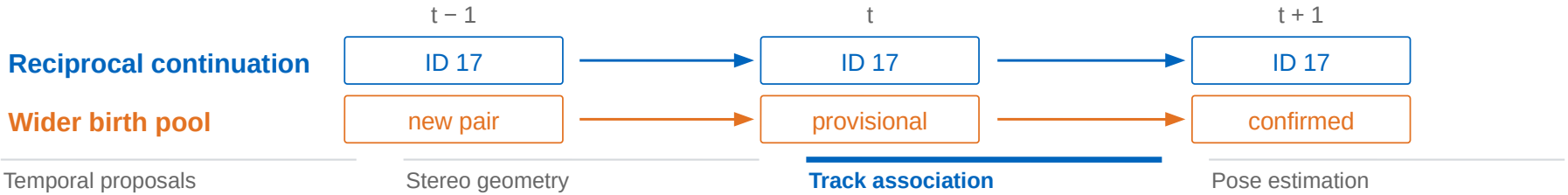


Accepted · frame 70



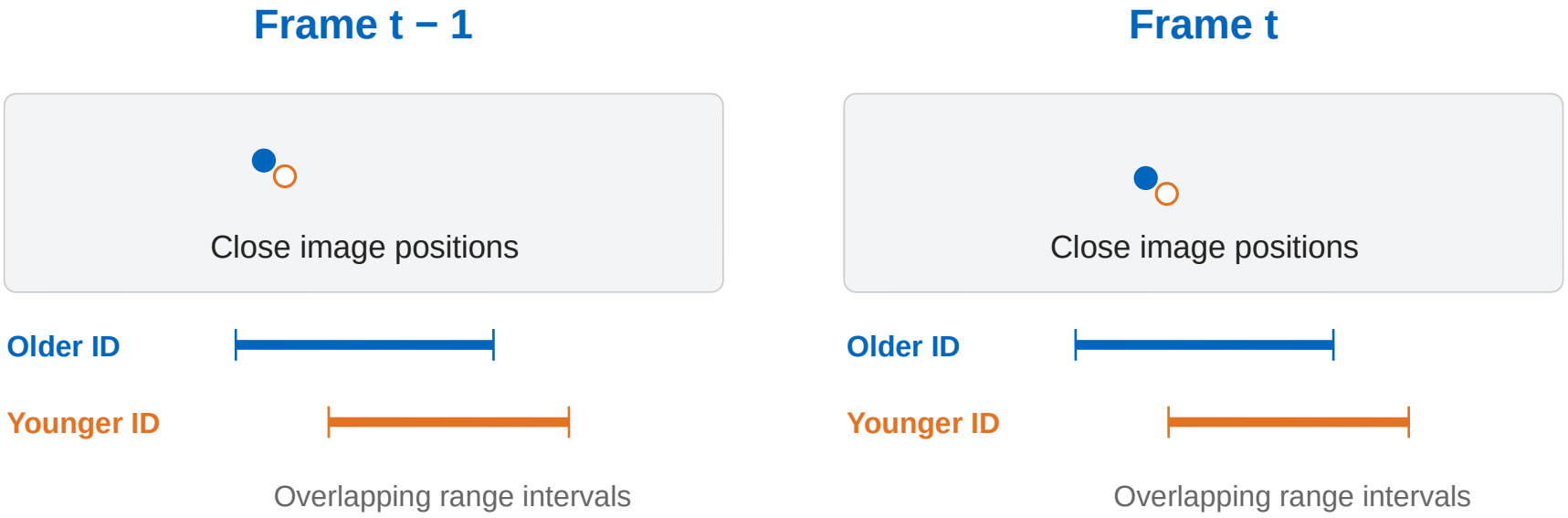
Rejected · frame 291

Cyan: KLT diagnostic · magenta: codec endpoint



Consensus / ID drawing: schematic. Measured crops: thesis Fig. 4.10, distinct frames. KLT markers are post-hoc diagnostics.

Preserve continuity without duplicating support

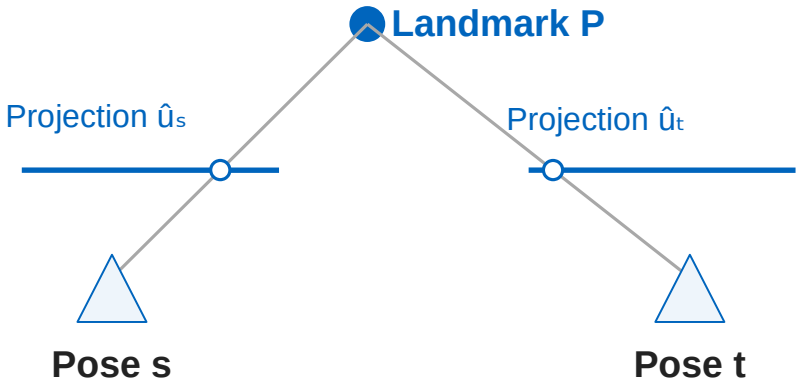


Retain the older ID when both endpoints are indistinguishable.

Schematic · thesis §4.8. Selected position tolerance: 1.125 px at both endpoints, with overlapping depth intervals.

Optimize the displacement supplied by the codec

One shared landmark; two camera poses



Predicted displacement

$$\hat{u}_t - \hat{u}_s$$

Measured endpoint difference

$$m_t = u_t - u_s$$

$$r = (\hat{u}_t - \hat{u}_s) - m_t$$

Optimize camera poses and landmark parameters jointly.

Schematic · thesis §4.9. Absolute factors remain for hosts and observations without an eligible predecessor.

Evaluate retained geometry on all eleven EuRoC sequences

Calibrated stereo · no IMU

Easy, medium, difficult
Full-sequence evaluation

ATE: global position error
One rigid alignment; no scale correction

RTE-6: relative translation error
Six frames, nominally 0.3 s

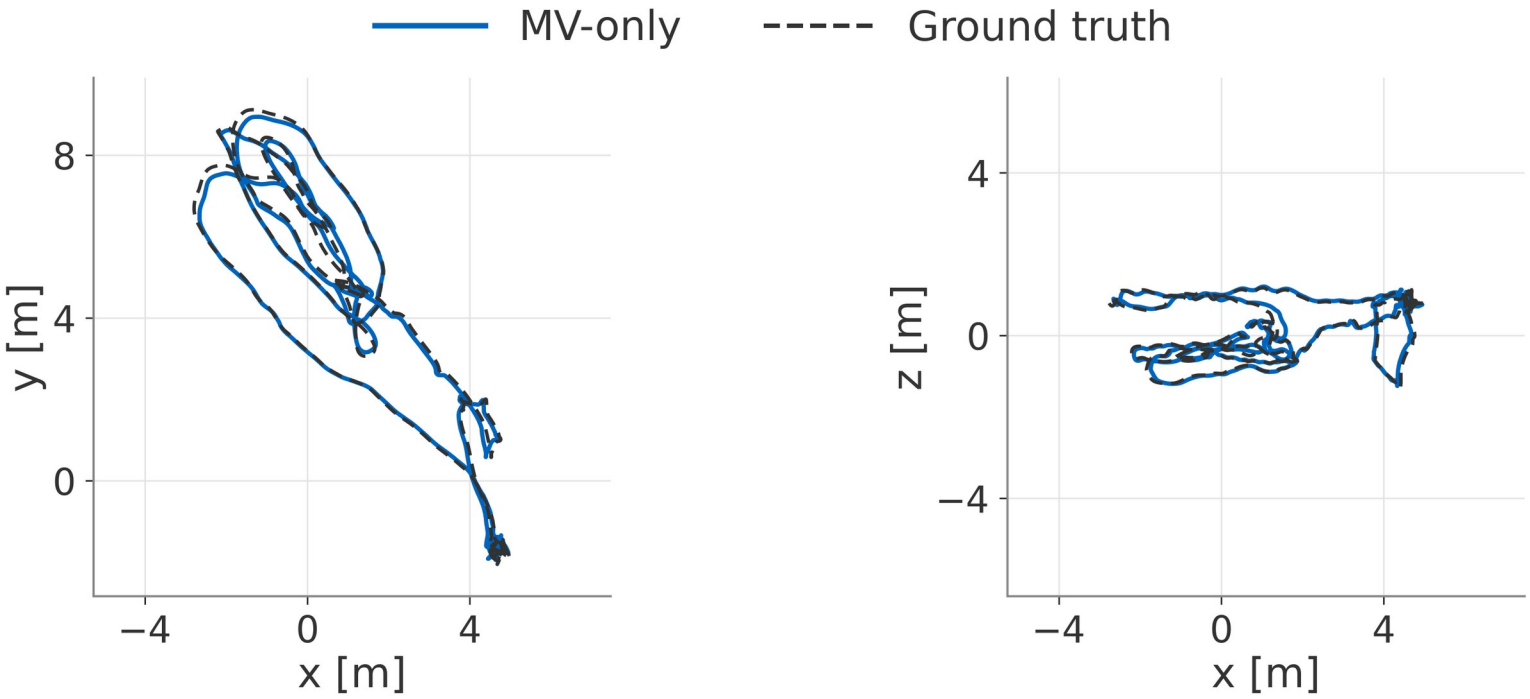
Method	Information used
PVO	Codec metadata + geometry
VOCA MV-first	MV-initialized KLT + retry
VOCA OF-first	Optical flow + MV retry

VOCA references retain pixel refinement.

Trajectory examples: lowest-ATE variant per sequence.

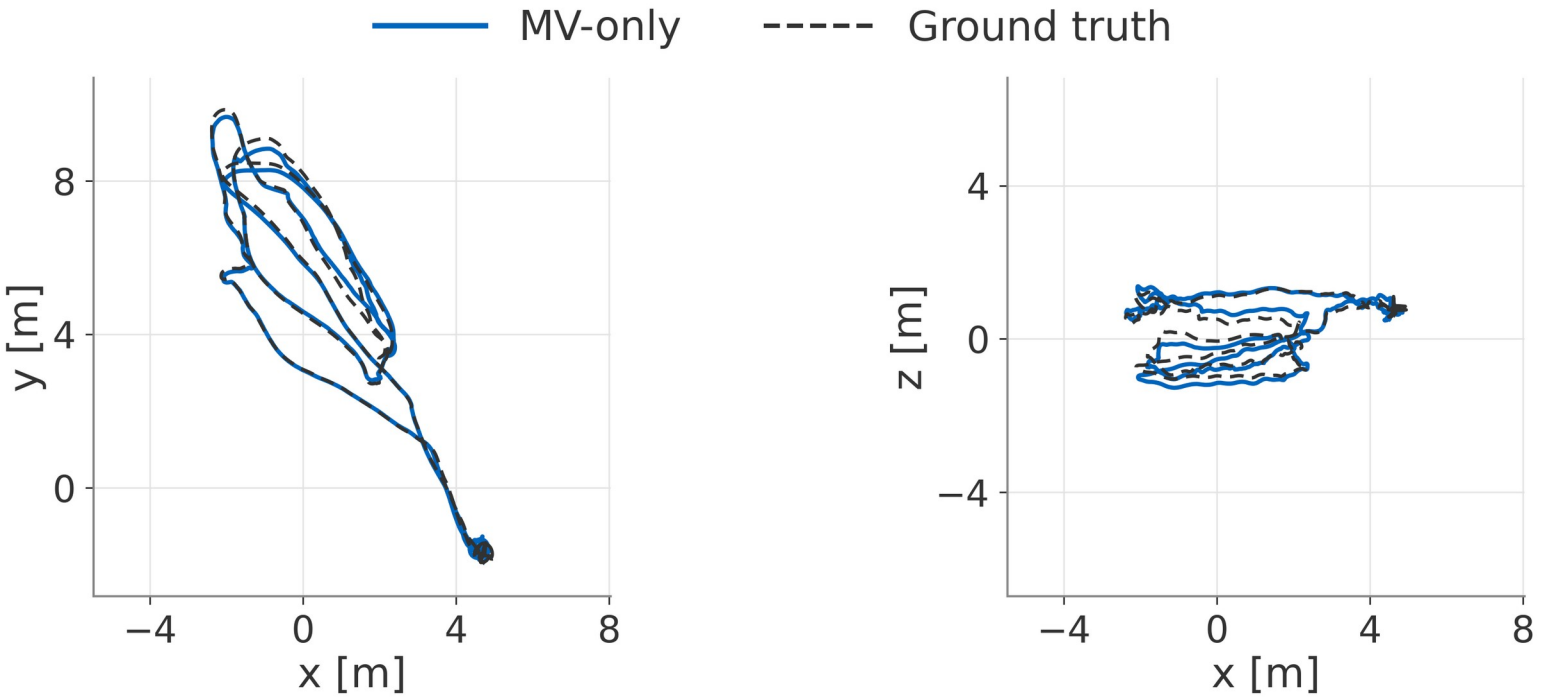
Thesis §§5.2–5.6; Tables 6.1–6.6. EuRoC [6], frozen VOCA [4]. Trajectory variants are identified in the notes.

EMH01: ATE 0.129 m



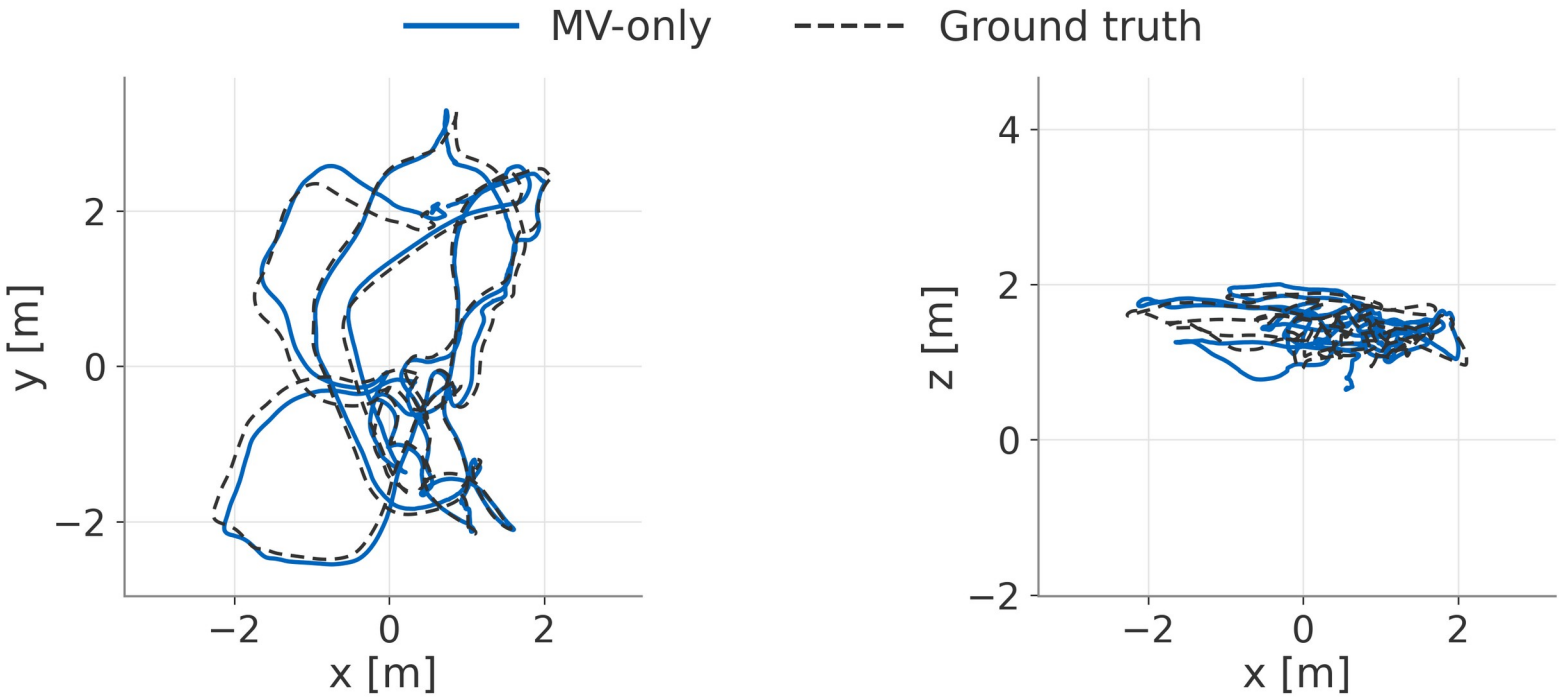
MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

EMH02: ATE 0.203 m



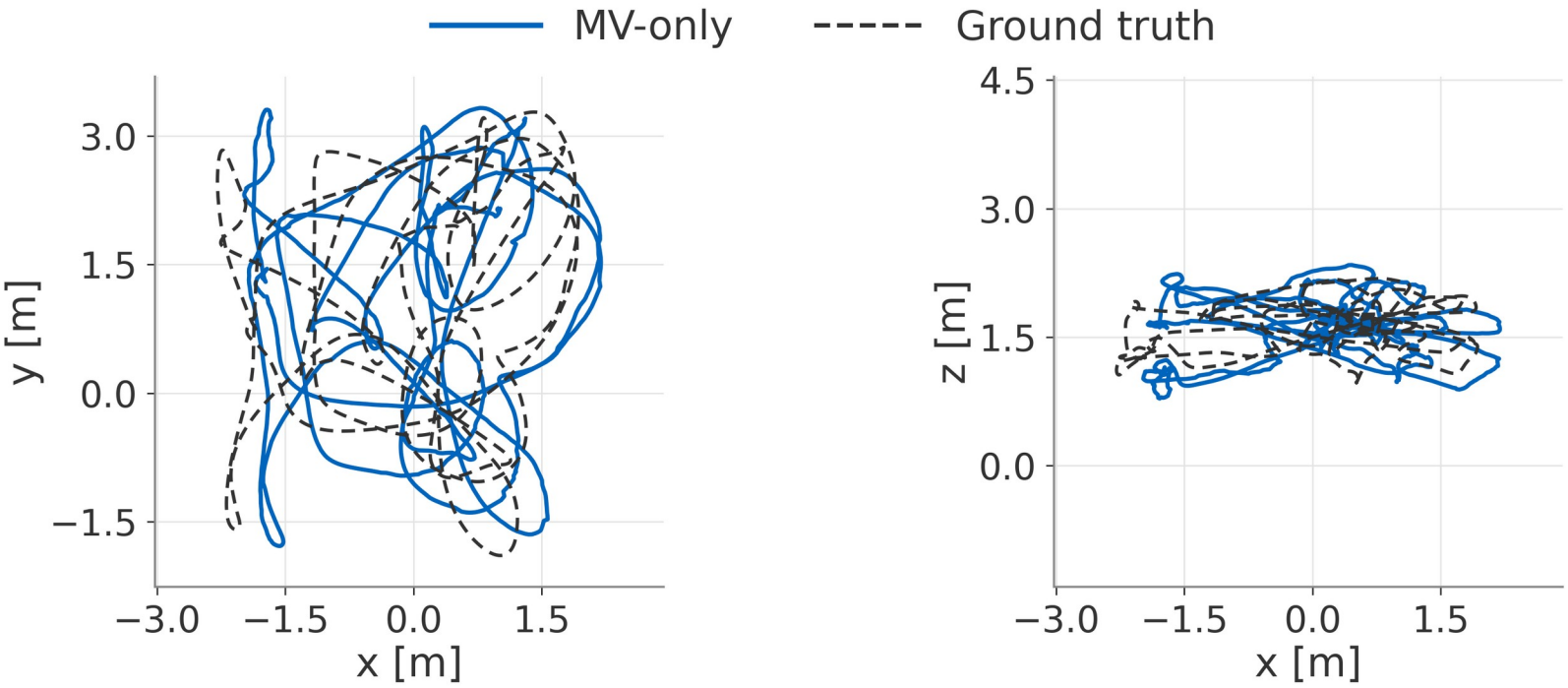
MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

EV101: ATE 0.223 m



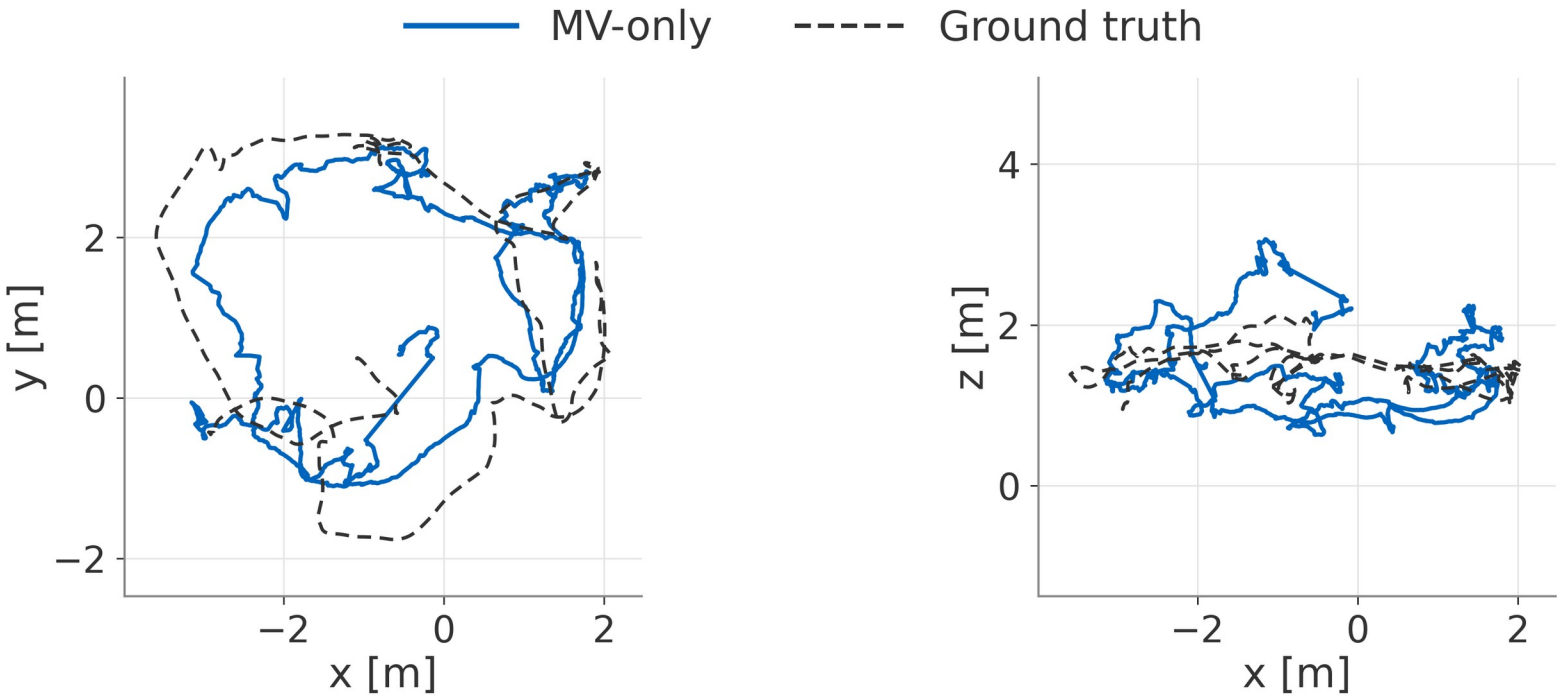
MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

EV102: ATE 0.501 m



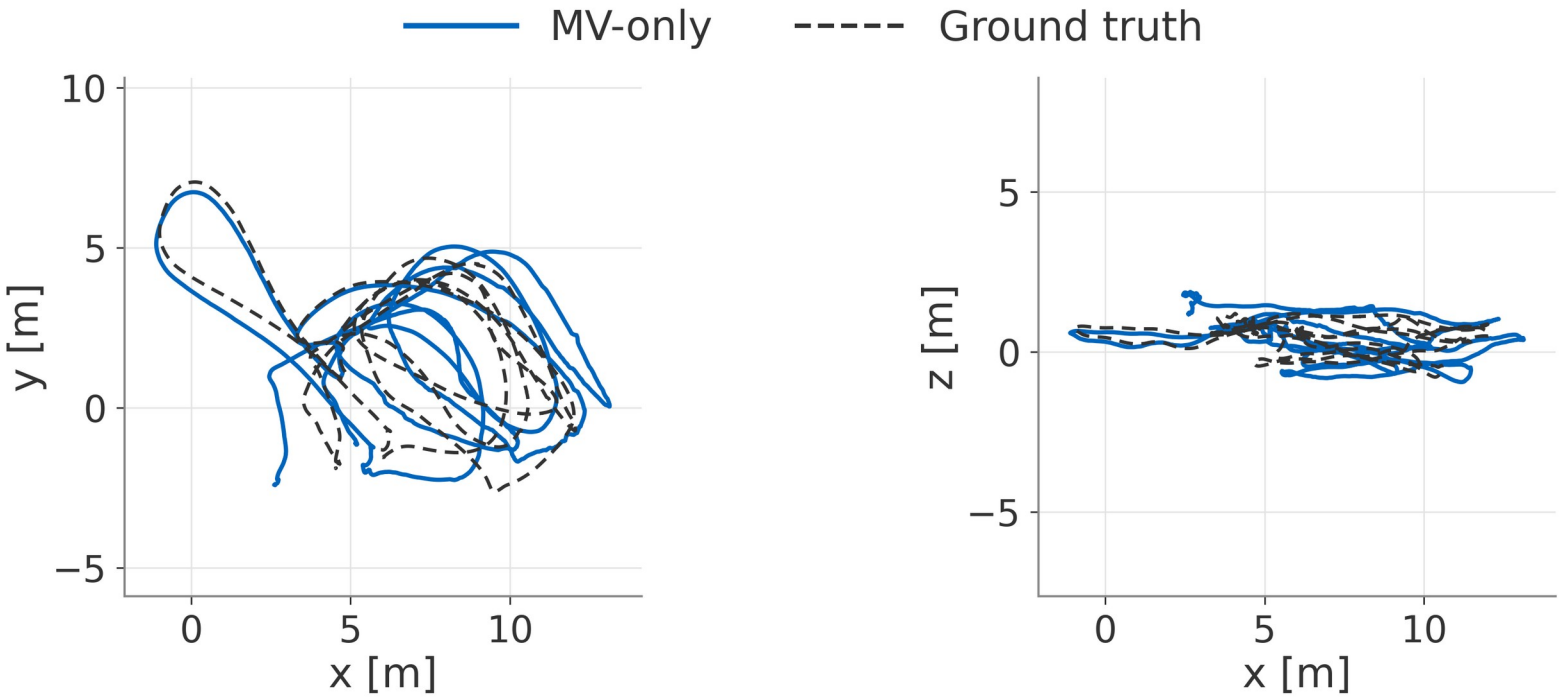
MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

EV201: ATE 0.727 m



MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

EMH03: ATE 1.082 m



MV-only · full sequence · fixed-scale rigid alignment. Thesis Table 6.4; variant and archive source in notes.

Selected PVO: retained accuracy across all eleven sequences

0.246 m

EMH01 ATE

0.226 m

EV101 ATE

0.0092 m

EMH01 RTE-6

PVO difficulty	Mean ATE [m]	Mean RTE-6 [m]
Easy	0.500	0.0306
Medium	1.228	0.0971
Difficult	6.968	0.7035

Easy-group RTE-6: 0.0101 m above VOCA MV-first on average.

Pixel-refined references generally remain more accurate.

Selected PVO · signed thesis Tables 6.1–6.3 and 7.1. Difficulty means include all eleven sequences and large failures.

Balance correspondence quality and coverage

Controlled frontend variants and the separately selected PVO operating point

ATE [m]	TEASER++ only	+ F–B	+ Suppression	Joint frontend	PVO
EMH01	0.129	0.488	0.143	0.232	**0.246**
EMH02	0.310	0.352	0.296	0.203	**0.594**
EV101	0.381	0.340	0.434	0.223	**0.226**
EV102	0.596	0.501	0.594	0.622	**0.514**
EV201	1.673	0.727	1.968	0.899	**0.935**
EMH03	3.261	1.559	1.508	1.569	**1.082**

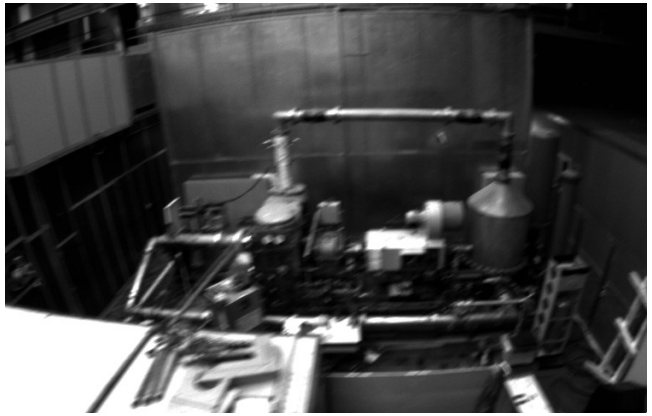
On EMH03, either added component improves the original baseline.

Selected PVO has the lowest medians; its large failures worsen the means.

Thesis Tables 6.4–6.5. TEASER++ + rigid flow are active throughout; selected PVO also changes thresholds and backend weights.

Visible texture must become usable temporal support

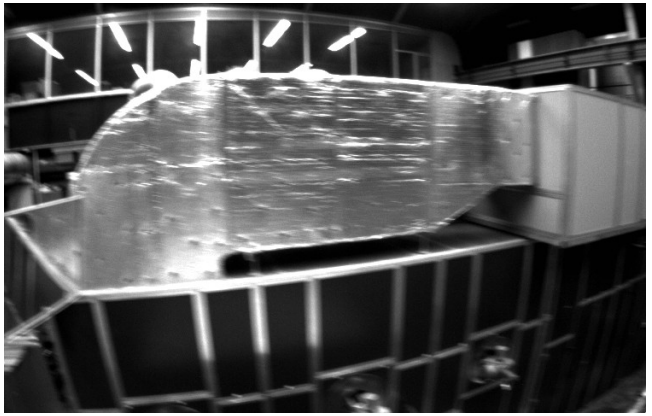
Error burst · 106.95 s



12.6 connected IDs/frame

RTE-6: 0.4423 m

Comparison · 67.05 s



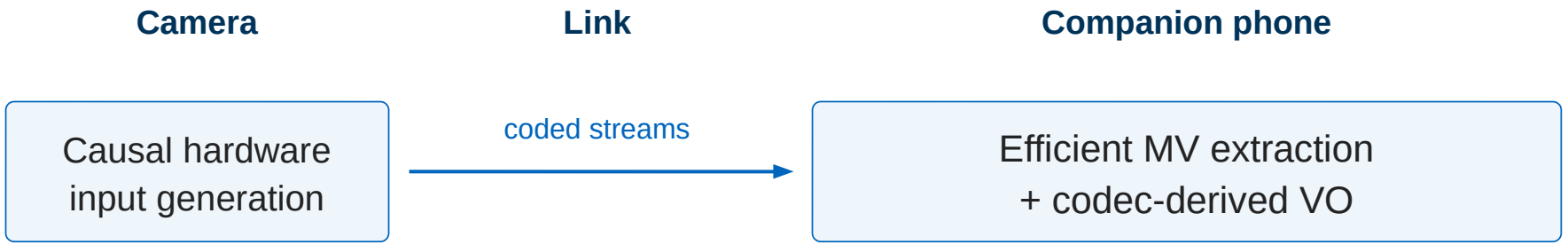
186.4 connected IDs/frame

RTE-6: 0.0265 m

Corners: 656 / 670. Speed: burst 2.51 m/s; comparison 2.66 m/s.

EMH03 diagnostic intervals · thesis §6.8.2 / Fig. 6.11. Counts are interval means; this comparison does not isolate one cause.

From geometric feasibility to a camera–phone system



Demonstrated: codec-only geometry and useful camera trajectories.

Next: causal hardware and transport, measured together for energy, latency, bandwidth, and tracking.

Total energy and latency are unmeasured; additional stereo encodings carry a cost.

Thesis §§7.6, 8.2. Coded streams are CTA; later transmission cuts are ATC-like / hybrid allocations with state requirements.

Video compression can also supply tracking geometry

1 **A concrete geometric construction**

Codec stereo → validated tracks → optimized trajectory

2 **Useful visual odometry**

Complete metric trajectories without pixel refinement

3 **A reason to reuse video processing**

Connect correspondence reuse with distributed analysis

Thank you.

Thesis §§1.4, 8.1. One demonstrated construction for further exploration.

Backup: why optimize displacement?

Paired experiment: change only the observation factor.

Across 11 sequences	Absolute reprojection	Rigid flow
Median ATE [m]	1.215	1.082
Median RTE-6 [m]	0.0824	0.0818
Mean ATE [m]	2.046	3.051
Mean RTE-6 [m]	0.1326	0.2934

Rigid flow lowers ATE on 8/11 sequences and RTE-6 on 7/11.

Large failures, particularly EV103, make the means favour absolute reprojection.

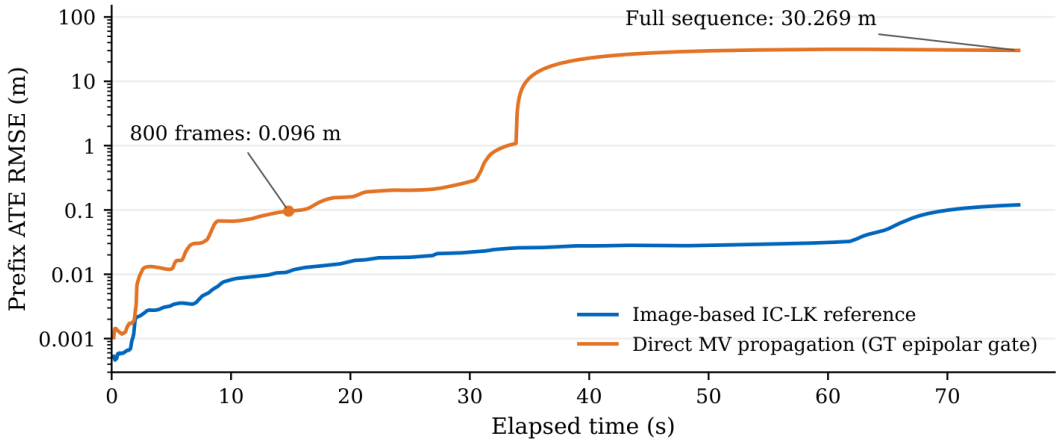
Signed thesis §6.4, Table 6.6. Full-sequence paired experiment; lower errors are better.

Backup: lessons from earlier attempts

Frequency-based selection sought geometrically consistent MVs.

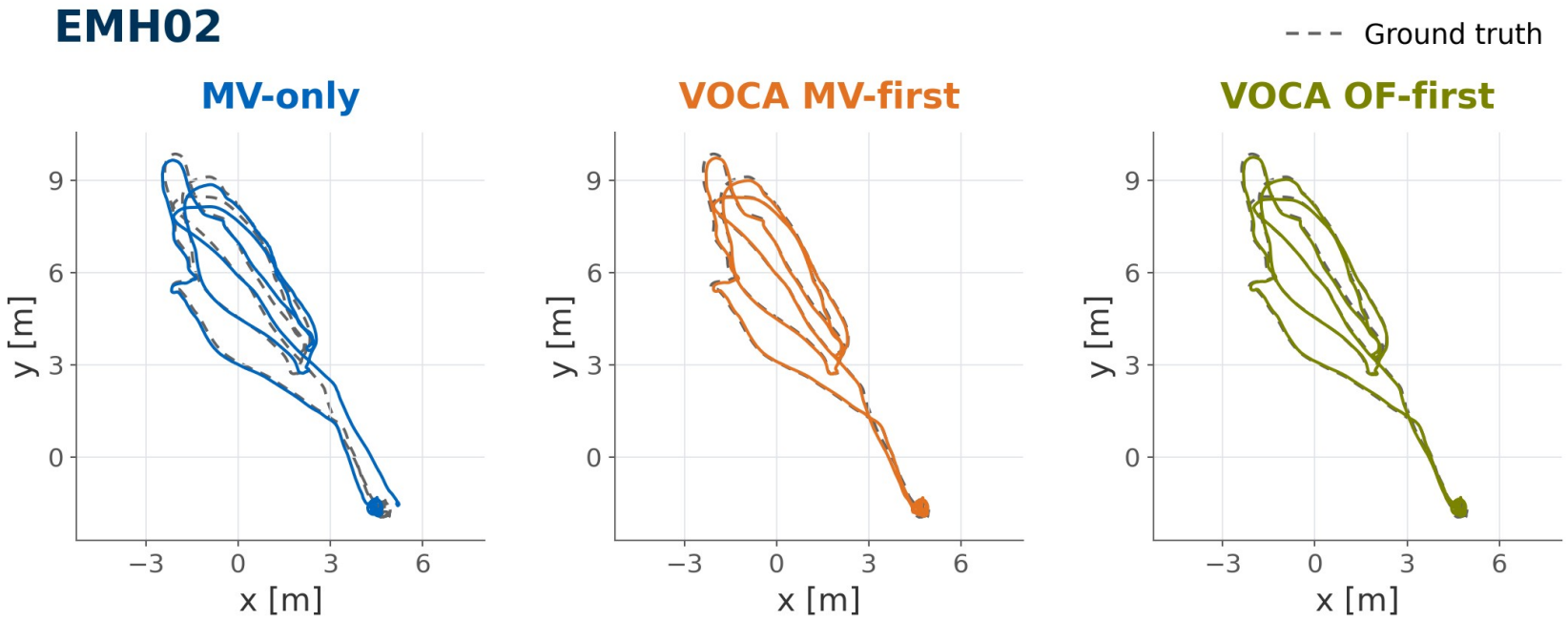
Direct feature propagation could look promising briefly, then diverge.

Useful motion and persistent identity are separate requirements.



Signed thesis Appendix B.4. Archived Monado MIO07 prototype; separate rigid alignment for each prefix.

Backup: EMH02 qualitative comparison



Thesis Fig. 6.3. MV-only curve: suppression variant; VOCA reference configurations shown separately.

Backup: complete EuRoC ATE results

ATE [m]	PVO	VOCA MV-first	VOCA OF-first
EMH01	**0.246**	0.172	0.118
EMH02	**0.594**	0.123	0.104
EMH03	**1.082**	0.200	0.212
EMH04	**4.722**	0.234	0.202
EMH05	**4.089**	0.228	0.181
EV101	**0.226**	0.101	0.096
EV102	**0.514**	0.074	0.085
EV103	**12.565**	0.513	0.438
EV201	**0.935**	0.267	0.090
EV202	**2.087**	0.137	0.104
EV203	**6.497**	9.532	2.508

Signed thesis Table 6.1. Full sequences; one rigid alignment, fixed scale; lower is better.

Backup: complete EuRoC RTE-6 results

RTE-6 [m]	PVO	VOCA MV-first	VOCA OF-first
EMH01	**0.0092**	0.0063	0.0066
EMH02	**0.0123**	0.0060	0.0061
EMH03	**0.0476**	0.0150	0.0145
EMH04	**0.1344**	0.0213	0.0196
EMH05	**0.1195**	0.0159	0.0162
EV101	**0.0189**	0.0142	0.0140
EV102	**0.0359**	0.0163	0.0155
EV103	**2.0434**	0.0618	0.0643
EV201	**0.0818**	0.0551	0.0163
EV202	**0.2078**	0.0167	0.0145
EV203	**0.5169**	1.5336	0.2455

Signed thesis Table 6.2. Relative translation RMSE over non-overlapping six-frame intervals; lower is better.

Backup: full component ablation — ATE

ATE [m]	TEASER++ only	F–B agreement	Suppression	Joint frontend	Selected PVO
EMH01	0.129	0.488	0.143	0.232	**0.246**
EMH02	0.310	0.352	0.296	0.203	**0.594**
EMH03	3.261	1.559	1.508	1.569	**1.082**
EMH04	3.527	6.045	4.856	5.672	**4.722**
EMH05	5.629	5.401	4.706	5.318	**4.089**
EV101	0.381	0.340	0.434	0.223	**0.226**
EV102	0.596	0.501	0.594	0.622	**0.514**
EV103	2.536	2.196	2.019	3.406	**12.565**
EV201	1.673	0.727	1.968	0.899	**0.935**
EV202	2.495	2.392	2.040	2.671	**2.087**
EV203	4.842	5.817	6.479	7.635	**6.497**
Macro mean	2.307	2.347	2.277	2.586	**3.051**
Median	2.495	1.559	1.968	1.569	**1.082**

Signed thesis Table 6.4. First four columns: original operating point. Selected PVO: separate parameter selection.

Backup: full component ablation — RTE-6

RTE-6 [m]	TEASER++ only	F–B agreement	Suppression	Joint frontend	Selected PVO
EMH01	0.0091	0.0118	0.0092	0.0115	**0.0092**
EMH02	0.0112	0.0155	0.0112	0.0159	**0.0123**
EMH03	0.0848	0.1025	0.0477	0.0978	**0.0476**
EMH04	0.1779	0.1767	0.1476	0.1488	**0.1344**
EMH05	0.1414	0.2290	0.1226	0.1365	**0.1195**
EV101	0.0363	0.0252	0.0343	0.0190	**0.0189**
EV102	0.0368	0.0480	0.0365	0.0494	**0.0359**
EV103	0.3264	0.2262	0.2839	0.2585	**2.0434**
EV201	0.2674	0.1134	0.4599	0.1005	**0.0818**
EV202	0.2328	0.2735	0.2102	0.2564	**0.2078**
EV203	0.5989	0.4523	0.7135	0.4365	**0.5169**
Macro mean	0.1748	0.1522	0.1888	0.1392	**0.2934**
Median	0.1414	0.1134	0.1226	0.1005	**0.0818**

Signed thesis Table 6.5. First four columns: original operating point. Selected PVO: separate parameter selection.

Backup: why the final evaluation uses EuRoC

Pilot setup	Native-to-rectified coverage	Depth at both temporal endpoints
EuRoC	97.0%	74.1–74.8%
TUM-VI	51.5%	about 43%
Monado	58.4–78.7%	6.4–17.3%

The implemented stereo lookup needs broad common image support.

Missing endpoint depth removes a candidate from 3D–3D association.

Signed thesis §5.2. Pilot configurations, not a general ranking of datasets. Dataset references [6, 8, 9].

Backup: deployment interfaces and efficiency

Possible transport cut	Allocation and required information
Coded streams	CTA: decode/extract and analyze at receiver
Extracted observations	Hybrid: support, references, calibration, timestamps
Confirmed frontend tracks	Hybrid: stable IDs, displacements and maintained state

Current preparation: 2 forward + 2 reverse temporal + 12 stereo streams.

Reported platform example: 80 mW video/audio hardware; 1000 mW CPU.

Total energy and latency remain unmeasured; extra encodings have costs.

Signed thesis §§1.1, 7.6–7.7. Chaurasia et al. [1]: platform budgets, not measured VO savings.

References: motivation and method

[1] Chaurasia et al. Passthrough+: Real-time Stereoscopic View Synthesis for Mobile Mixed Reality. PACM CGIT 3(1), 2020.

[2] Hofer et al. H.264 Compress-Then-Analyze Transmission in Edge-Assisted Visual SLAM. European Wireless, pp. 130–135, 2023.

[3] Redondi et al. Compress-then-Analyze versus Analyze-then-Compress: What Is Best in Visual Sensor Networks? IEEE TMC 15(12), pp. 3000–3013, 2016.

[4] Hilscher et al. VOCA: Visual Odometry with Codec Awareness. ECCV, 2026.

[5] Yang, Shi, and Carlone. TEASER: Fast and Certifiable Point Cloud Registration. IEEE T-RO 37(2), pp. 314–333, 2021.

Bibliographic records follow the signed thesis; numbering is local to this presentation.

References: evaluation and future directions

[6] Burri et al. The EuRoC Micro Aerial Vehicle Datasets. IJRR 35(10), pp. 1157–1163, 2016.

[7] Li and Song. Toward Featureless Visual Navigation: Simultaneous Localization and Planar Surface Extraction Using Motion Vectors in Video Streams. ICRA, pp. 9–14, 2014.

[8] Schubert et al. The TUM VI Benchmark for Evaluating Visual-Inertial Odometry. IROS, 2018.

[9] de Mayo, Cremers, and Pire. The Monado SLAM Dataset for Egocentric Visual-Inertial Tracking. IROS, pp. 13111–13118, 2025.

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